#HW 12

Graded

Student

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Total Points

100 / 100 pts

Question 1

(no title) **30** / 30 pts

✔ **– 0 pts** Correct

Question 2

(no title) **10** / 10 pts

✔ **– 0 pts** Correct

– 1 pt minimum theoreotical power input value?

– 1 pt wrong Tc and Th temp. taken

– 1 pt is 3 hp sufficient or not?

Question 3

(no title) **15** / 15 pts

✔ **– 0 pts** Correct

– 3 pts reversible heat pump calculation?

– 5 pts incomplete solution

Question 4

(no title) **15** / 15 pts

✔ **– 0 pts** Correct

– 2 pts Pressure at end of isothermal expansion = 76.1 kPa

– 4 pts pressure not calculated

Question 5

(no title) **15** / 15 pts

✔ **– 0 pts** Correct

– 2 pts Case (a) is irreversible

Question 6

(no title) **15** / 15 pts

✔ **− 0 pts** Correct

− 3 pts incomplete solution

Questions assigned to the following page: 2 and 3

2) 6.17.45

$T_C = 70°F = 70 + 460 = 530°R$

$T_H = 90°F = 90 + 460 = 550°R$

$$COP_{max} = \frac{T_C}{T_H - T_C} = \frac{530°R}{550°R - 530°R} = 26.5$$

$$\dot{W}_{min} = \frac{\dot{Q}_{in}}{COP_{max}} = \frac{30000\ Btu/h}{26.5} \approx 1132.08\ Btu/h$$

$$\dot{W}_{min,hp} = \frac{1132.08\ Btu/h}{2544.43\ Btu/h/hp} \approx 0.445\ hp$$

The proposed input is 3hp. Since the minimum theoretical power required (0.445hp) is less than the available power (3hp), the power input of 3hp is sufficient

$$COP_{actual} = \frac{\dot{Q}_{in}}{\dot{W}_{actual}}$$

$$\dot{W}_{actual} = 3hp \times 2544.43\ Btu/h/hp \approx 7633.3\ Btu/h$$

$$\Rightarrow COP_{actual} = \frac{30000\ Btu/h}{7633.3\ Btu/h} = 3.93$$

⇒ Yes, a net power input of 3hp is sufficient because the minimum theoretical power required is approximately 0.445 hp

The coefficient of performance is approximately 3.93

3) 6.17.47

a) $1kW = 56.87\ Btu/min$

$$P_a = \frac{Q_H}{56.87\ Btu/min/kW} = \frac{740\ Btu/min}{56.87\ Btu/min/kW} = 13.01$$

$$\Rightarrow P_{input,a} = P_a = 13.01\ kW$$

b) $$COP = \frac{Q_H}{P_{input,b}} \Rightarrow P_{input,b} = \frac{Q_H}{COP} = \frac{13.01\ kW}{3.0} = 4.34\ kW$$

c) $T_H = 68°F = 68 + 459.67 = 527.67°R$

$T_C = 32°F = 32 + 459.67 = 491.67°R$

Questions assigned to the following page: 3, 4, and 5

$$COP_{rev} = \frac{T_H}{T_H - T_C} = \frac{527.67°R}{527.67°R - 491.67°R} = 14.66$$

$$\dot{W}_{input,C} = \frac{Q_H}{COP_{rev}} = \frac{13.01\,kW}{14.66} = 0.887\,kW$$

4) 6.17.53

a) $$COP = \frac{T_H}{T_H - T_C} = \frac{600K}{600K - 300K} = 2$$

$$COP = \frac{|\dot{Q}_H|}{|\dot{W}_{net}|} \Rightarrow |\dot{W}_{net}| = \frac{|\dot{Q}_H|}{COP} = \frac{250\,kJ/kg}{2} = 125\,kJ/kg$$

b) $p_4 v_4 = p_3 v_3$

$$Q_H = RT_H \ln\left(\frac{v_3}{v_4}\right)$$

$$\Rightarrow 250 = 0.287 \cdot 600 \ln\left(\frac{v_3}{v_4}\right) \Rightarrow \ln\left(\frac{v_3}{v_4}\right) = \frac{250}{172.2} = 1.452 \Rightarrow \frac{v_3}{v_4} = 4.27$$

isothermal $\Rightarrow$ $$p_3 = p_4 \frac{v_4}{v_3} = \frac{325}{4.27} = 76.1\,kPa$$

5) 6.17.58

a) $\dot{Q}_H = 600\,kW$ $\quad \dot{Q}_C = 400\,kW$

$$W_{net} = \dot{Q}_H - \dot{Q}_C = 600 - 400\,kW = 200\,kW$$

$$\eta = \frac{W_{net}}{\dot{Q}_H} = \frac{200}{600} = 0.333$$

$\Rightarrow \eta < \eta_{max}$ $(0.333 < 0.6667)$ $\Rightarrow$ irreversible

b) $\dot{Q}_H = 600\,kW$, $\dot{Q}_C = 0$

$W_{net} = 600 - 0 = 600\,kW$

$$\eta = \frac{600}{600} = 1 = 100\%$$

$\Rightarrow \eta > \eta_{max}$ $(1 > 0.6667)$ $\Rightarrow$ impossible

Questions assigned to the following page: 5 and 6

c) $\dot{Q}_H = 600\,kW \qquad \dot{Q}_C = 200\,kW$

$W_{net} = 600 - 200 = 400\,kW$

$\eta = \frac{400}{600} = 0.6667$

$\Rightarrow \eta = \eta_{max} \; (0.6667 = 0.6667) \Rightarrow$ Reversible

6) 6.17.63

$$\sigma_{cycle} = \sum \frac{Q_k}{T_k}$$

$$= \frac{750}{1500} + \frac{-100}{500} + \frac{Q_3}{1000}$$

$$\Rightarrow \sigma_{cycle} = 0.3 + \frac{Q_3}{1000}$$

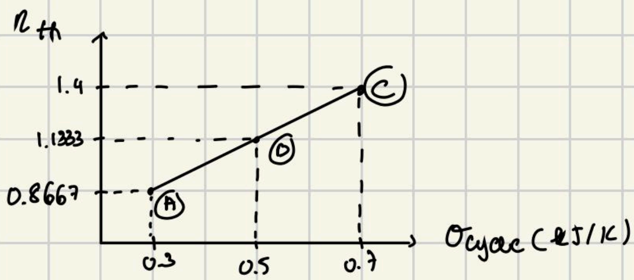
$$\Rightarrow Q_3 = 1000(\sigma_{cycle} - 0.3)$$

$$\eta = \frac{W}{Q_{in}} = \frac{W}{750}$$

$$W = Q_1 + Q_2 + Q_3 = 750 - 100 + Q_3 = 650 + Q_3$$

$$\Rightarrow \eta(\sigma) = \frac{650 + 1000(\sigma - 0.3)}{750} = \frac{650 + 1000\sigma - 300}{750} = \frac{350 + 1000\sigma}{750}$$

$$\Rightarrow \eta(\sigma) = 1.333\sigma + 0.4667$$



$\eta(\sigma) = 1.333\sigma + 0.4667$

A(0.3, 0.8667)

B(0.5, 1.1333)

C(0.7, 1.4)

Question assigned to the following page: 1

1)

HW 12 Question 1

52 / 52 points

Due: 11/25/2025, 11:59 PM CST

4.1 Retrieving entropy data	7 / 7 points
4.2 Introducing the T dS equations	4 / 4 points
4.3 Entropy - a system property	3 / 3 points
4.4 Entropy change of an ideal gas	4 / 4 points
4.5 Entropy change in internally reversible processes of closed systems	6 / 6 points
4.6 Entropy change of an incompressible substance	4 / 4 points
4.7 Entropy balance for closed systems	7 / 7 points
4.8 Directionality of processes	7 / 7 points
4.9 Entropy rate balance for control volumes	4 / 4 points
4.10 Rate balances for control volumes at steady state	6 / 6 points